```
#include <iostream>
using namespace std;

class node
{
public:
   int data;
   node *next;

   node(int x)
   {
      data = x;
      next = NULL;
   }
};

class queue
{
private:
   node *front;
   node *rear;

public:
   queue()
   {
      front = rear = NULL;
   }

   int empty(void)
   {
      if (front == NULL)
      {
         return 1;
      }
      else
      {
         return 0;
      }
   }

   int peek(void)
   {
      if (empty())
      {
         cout << "\nQueue is empty";
         return -1;
      }
```

```
    return front->data;
}

void enqueue(int x)
{
    node *p = new node(x);

    if (front == NULL)
    {
        front = p;
        rear = p;
    }
    else
    {
        rear->next = p;
        rear = p;
    }
}

int dequeue(void)
{
    if (empty())
    {
        cout << "\nQueue is empty";
        return -1;
    }

    int x = front->data;
    node *p = front;

    front = front->next;

    if (front == NULL)
    {
        rear = NULL;
    }

    delete p;

    return x;
}

void display(void)
{
    node *q = front;

    cout << "\nQueue elements: ";
```

```
        while (q != NULL)
        {
            cout << "\t" << q->data;
            q = q->next;
        }
    }
};

int main()
{
    int x;
    queue q;

    q.enqueue(27);
    q.enqueue(34);
    q.enqueue(42);
    q.enqueue(58);
    q.enqueue(65);
    q.enqueue(70);

    q.display();

    if (!q.empty())
    {
        x = q.dequeue();
        cout << "\nRemoved: " << x;
    }

    q.display();

    x = q.peek();
    cout << "\nFront element is: " << x;

    return 0;
}
/*output:
PS C:\Users\Students> cd "d:\yadnya abhale\" ; if ($?) { g++
queueusing_linkedlist.cpp -o queueusing_linkedlist } ; if ($?) { .
\queueusing_linkedlist }

Queue elements:         27      34      42      58      65      70
Removed: 27
Queue elements:         34      42      58      65      70
Front element is: 34
PS D:\yadnya abhale> */
```